



10310 - Unveiling the Missing Sulfur: JWST Mapping of Atomic Sulfur in Protostellar Environments

Cycle: 5, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

JWST programs #2151, #1802 and #5804 continue to provide transformative insights into the chemistry and physics of protostellar environments. Notably, towards most of the targets from these programs, we identify a resolved and pronounced [S I] emission feature at 25.249 μm – unprecedented in S/N compared to observations from previous facilities. This feature is significant because it enables quantitative constraints on how much sulfur is locked in grains in dense interstellar clouds, providing critical insight into the long-standing ‘interstellar sulfur depletion’ problem. Yet, no study has systematically mapped the [S I] emission feature across a diverse range of protostars with sufficient accuracy to resolve sulfur partitioning in dense interstellar clouds, leaving a major aspect of the interstellar ‘missing sulfur’ problem unquantified.

We propose to map and measure the atomic sulfur abundance across >20 protostellar environments using data from JWST programs #2151, #1802 and #5804 to rigorously quantify the role of dense interstellar cloud grains as reservoirs of the ‘missing sulfur’ in dense regions of the interstellar medium. This work will additionally link sulfur partitioning to protostellar outflow dynamics and the distribution of Complex Organic Molecules,

adding new insights into the physico-chemical dynamics relevant to a wide range of JWST protostellar programs beyond this study (e.g., JOYS #1290). Having robust quantitative constraints on the partitioning of S in dense ISM grains, which feed into planet-forming material, will be essential for establishing the baseline inventory of sulfur available to emerging planets.

OBSERVING DESCRIPTION